

# Unit 7
# Building the IS-LM Model

Instructor: Nguyen Tai Vuong
School of Economics and Management
Hanoi University of Science and Technology



## *Learning Objectives*

- the IS curve, and its relation to the Keynesian cross
- the LM curve, and its relation to the theory of liquidity preference
- how the IS-LM model determines income/output and the interest rate in the short run when P is fixed

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## Contents

1. The IS curve
2. The LM curve
3. The Equilibrium in goods and money markets
4. Impact of fiscal and monetary policy under IS-LM Model





## The IS – LM Model

- IS–LM model, or Hicks–Hansen model, is a two-dimensional macroeconomic tool that shows the relationship between interest rates and real output in goods and services market.
  - The model was created, developed and taught by J. M. Keynes. However, it is often believed that John Hicks invented it in 1937, and was later extended by Alvin Hansen.
- IS: Investment – Saving
- LM: Liquidity preference – Money supply



# Building the IS Curve

- IS curve  Relationship between aggregate output and the interest rate in the goods market.
- The IS curve represents various combinations of interest and income/output along which the goods market is in equilibrium.



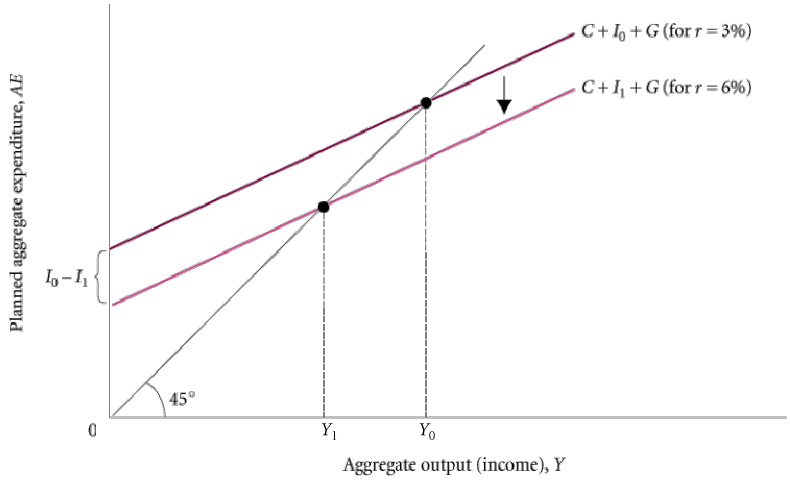
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# The Effect of an Interest Rate Change on Planned Aggregate Expenditure and Equilibrium Output

- A high interest rate (r) discourages planned investment (I).
- Planned investment is a part of planned aggregate expenditure (AE).
- Thus, when the interest rate rises, planned aggregate expenditure (AE) at every level of income falls.
- Finally, a decrease in planned aggregate expenditure lowers equilibrium output (income) (Y) by a multiple of the initial decrease in planned investment.

## The Effect of an Interest Rate Increase on Planned Aggregate Expenditure and Equilibrium Output




An increase in the interest rate from 3 percent to 6 percent lowers planned aggregate expenditure and thus reduces equilibrium output from $Y_0$ to $Y_1$.

## The Effect of an Interest Rate Change on Planned Aggregate Expenditure and Equilibrium Output

- Using a convenient shorthand:

$$r \uparrow \rightarrow I \downarrow \rightarrow AE \downarrow \rightarrow Y \downarrow$$

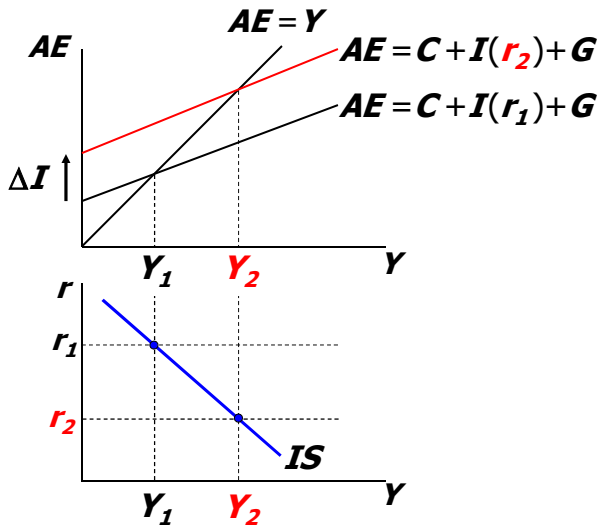
$$r \downarrow \rightarrow I \uparrow \rightarrow AE \uparrow \rightarrow Y \uparrow$$

# Deriving the IS curve

$\downarrow r \Rightarrow \uparrow I$

$\Rightarrow \uparrow AE$

$\Rightarrow \uparrow Y$




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# Why the IS curve is negatively sloped?

- A fall in the interest rate motivates firms to increase investment spending, which drives up total planned spending (AE).
- To restore equilibrium in the goods market, output (a.k.a. actual expenditure, Y) must increase.

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# Construct IS Function

Components of Keynesian Cross Model:

$$C = C_0 + mpcY_D$$
$$I = I_0 + I_r r$$
$$G = G_0$$
$$T = T_0$$

$$AE = C_0 + mpc(Y - T_0) + I_0 + I_r r + G_0$$
$$AE = [C_0 + I_0 + G_0 - mpc\ T_0] + mpcY + I_r r$$

At equilbirum in good market: $Y = AE$

→ IS: $Y = \frac{[C_0 + I_0 + G_0 - mpcT_0] + I_r r}{1 - mpc}$

An economy:
Good Market
$C = 200 + 0.8\ (Y - T)$
$I = 100 - 500r$
$G = 80$
$T = 80$
Find the IS function?





# Fiscal Policy and the IS curve

- We can use the IS-LM model to see how fiscal policy (G and T) affects aggregate demand and output.
- Let's start by using the Keynesian cross to see how fiscal policy shifts the IS curve…

# Shifting the IS curve: ΔG

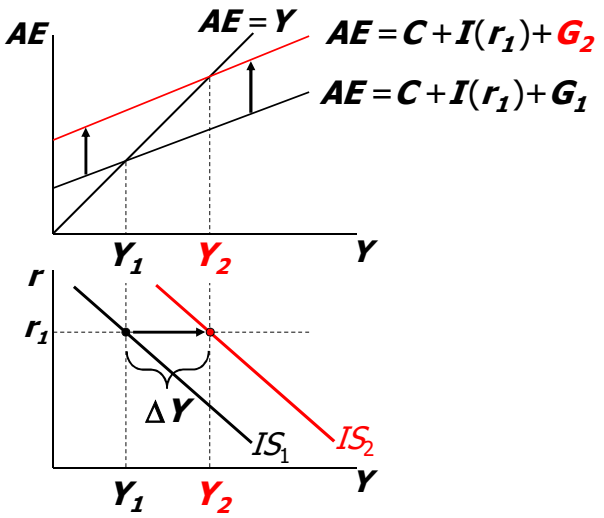
At any value of $r$,

$\uparrow G \Rightarrow \uparrow AE \Rightarrow \uparrow Y$

…so the *IS* curve shifts to the right.

The horizontal distance of the *IS* shift equals

$$\Delta Y = \frac{1}{1-\text{MPC}} \Delta G$$




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# Exercise: Shifting the IS curve

- Use the diagram of the Keynesian cross to show how an increase in taxes shifts the IS curve.

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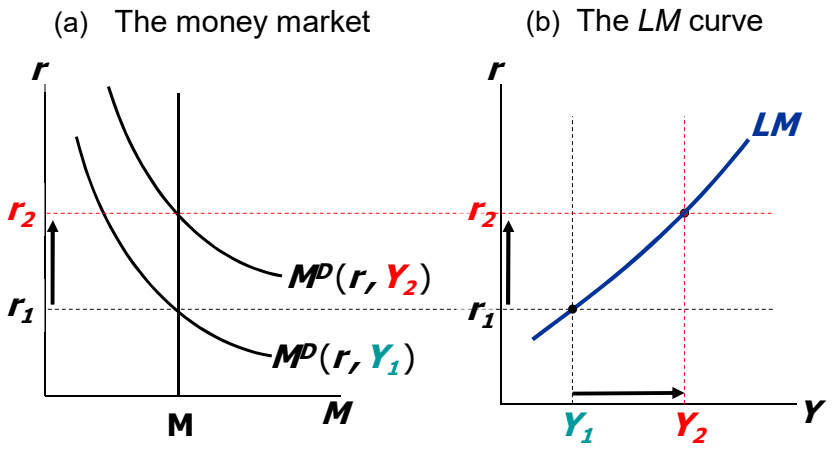
# Developing LM curve

- The LM curve shows the combinations of interest rates and levels of real income for which the money market is in equilibrium.

- At $Y_1$ → $M_1^D = C_M - c_r r + c_Y Y_1 \rightarrow M^S = M_1^D \rightarrow equilbrium\ r_1$
- At $Y_2$ → $M_2^D = C_M - c_r r + c_Y Y_2 \rightarrow M^S = M_2^D \rightarrow equilbrium\ r_2$
- $Y_2 > Y_1$ .....then ..... $r_2 > r_1$



# Deriving the *LM* curve





# Why the *LM* curve is upward sloping?

- An increase in income raises money demand.
- Since the supply of money is fixed, there is now excess demand in the money market at the initial interest rate.
- The interest rate must rise to restore equilibrium in the money market.

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# Contruct LM Function

- $\mathrm{M^D} = C_M - c_r r + c_Y Y$
- $\mathrm{M^S} = \mathrm{M1}$
- $\mathrm{M^S} = \mathrm{M^D} \rightarrow \mathrm{M1} = C_M - c_r r + c_Y Y$

$\rightarrow$ LM: $\mathrm{r} = \frac{(C_M - M1) + c_Y Y}{c_r}$

An economy:
Money market
$\mathrm{MD} = 200 - 300\mathrm{r} + 0.3\mathrm{Y}$
$\mathrm{MS} = 650$
Find the LM function?

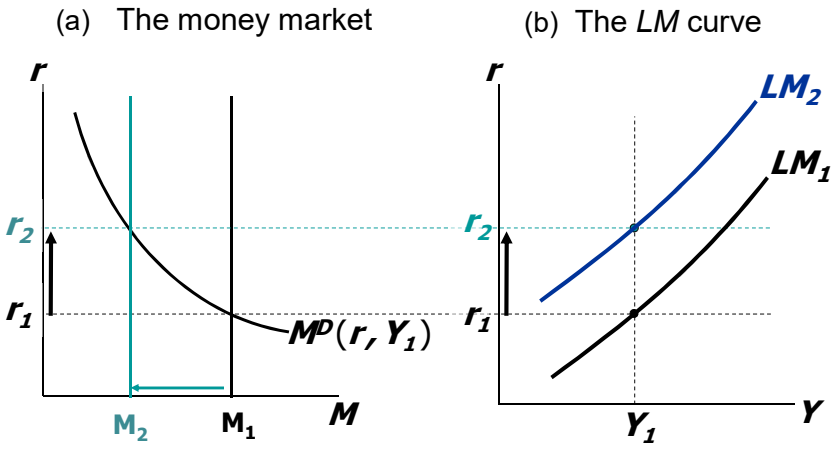


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# How $\Delta M$ shifts the *LM* curve



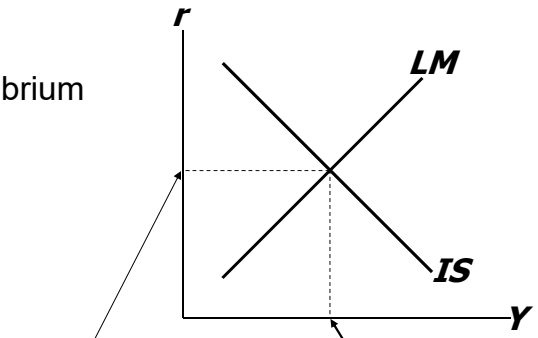



# The short-run equilibrium

The short-run equilibrium is the combination of ***r*** and ***Y*** that simultaneously satisfies the equilibrium conditions in the goods & money markets:

$\mathrm{Y} = \mathrm{AE} \rightarrow \mathrm{Y} = \mathrm{C} + \mathrm{I(r)} + \mathrm{G}$

$\mathrm{M^S} = \mathrm{M^D} \rightarrow \mathrm{M^S} = C_M - c_r r + c_Y Y$




Equilibrium interest rate

Equilibrium level of income

# Find Equilibrium Interest Rate and Income?

Good Market
$C = 200 + 0.8\ (Y - T)$
$I = 100 - 500r$
$G = 80$
$T = 80$
IS: ……

Money market
$MD = 200 - 300r + 0.3Y$
$MS = 650$
LM: ……

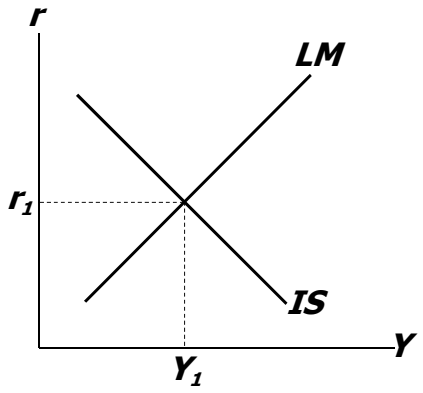




# Policy analysis with the *IS-LM* model

We can use the *IS-LM* model to analyze the effects of

- fiscal policy: ***G*** and/or ***T***
- monetary policy: ***M***







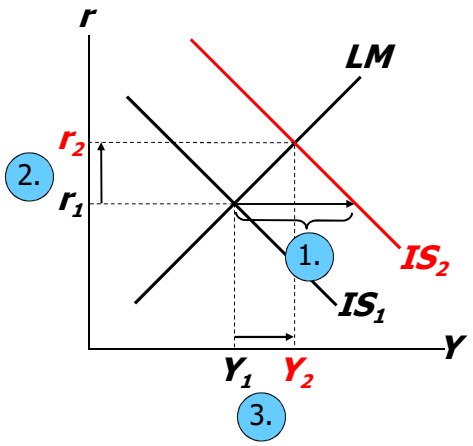

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# An increase in government purchases

1. *IS* curve shifts right
   by $\frac{1}{1-\text{MPC}}\Delta \boldsymbol{G}$
   causing output & income to rise.

2. This raises money demand, causing the interest rate to rise…

3. …which reduces investment, so the final increase in $\boldsymbol{Y}$ is smaller than $\frac{1}{1-\text{MPC}}\Delta \boldsymbol{G}$



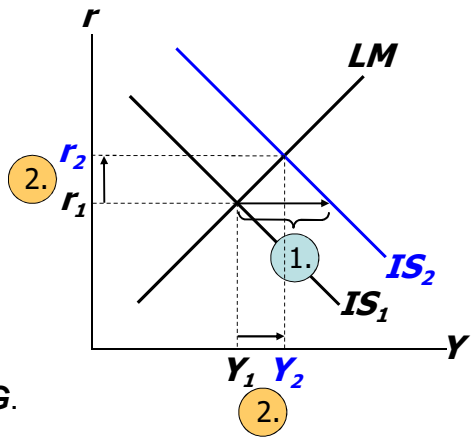



# A tax cut

Consumers save (1–*MPC*) of the tax cut, so the initial boost in spending is smaller for $\Delta \boldsymbol{T}$ than for an equal $\Delta \boldsymbol{G}$… and the *IS* curve shifts by

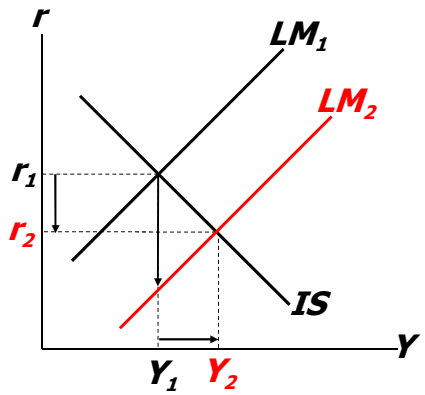
1. $\frac{-\text{MPC}}{1-\text{MPC}}\Delta \boldsymbol{T}$

2. …so the effects on $\boldsymbol{r}$ and $\boldsymbol{Y}$ are smaller for $\Delta \boldsymbol{T}$ than for an equal $\Delta \boldsymbol{G}$.



# Monetary policy: An increase in *M*

1. $\Delta M > 0$ shifts the *LM* curve down (or to the right)
2. …causing the interest rate to fall
3. …which increases investment, causing output & income to rise.




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# Interaction between monetary & fiscal policy

- Model:
  Monetary & fiscal policy variables (***M***, ***G,*** and ***T***) are exogenous.
- Real world:
  Monetary policymakers may adjust ***M*** in response to changes in fiscal policy, or vice versa.
- Such interaction may alter the impact of the original policy change.

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# The Central Bank's response to $\Delta G > 0$

- Suppose the government increases ***G***.
- Possible central bank responses:
  1. hold ***M*** constant
  2. hold ***r*** constant
  3. hold ***Y*** constant
- In each case, the effects of the $\Delta G$ are different:



# Response 1: Hold *M* constant

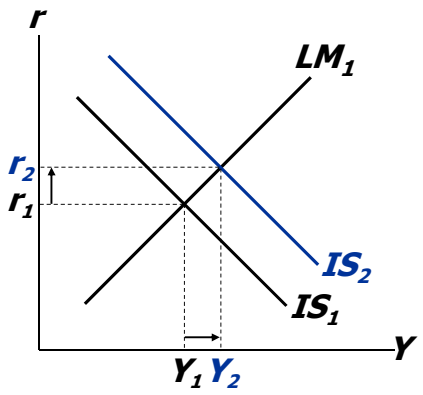
If the government raises ***G***, the *IS* curve shifts right.

If central bank holds ***M*** constant, then *LM* curve doesn't shift.

Results:

$$\Delta Y = Y_2 - Y_1$$

$$\Delta r = r_2 - r_1$$



# Response 2: Hold *r* constant

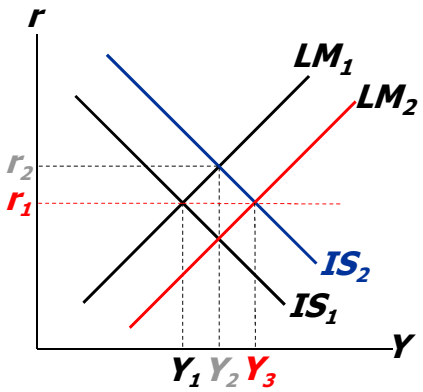
If the government raises ***G***, the *IS* curve shifts right.

To keep ***r*** constant, central bank increases ***M*** to shift *LM* curve right.

Results:

$$\Delta Y = Y_3 - Y_1$$

$$\Delta r = 0$$




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# Response 3: Hold *Y* constant

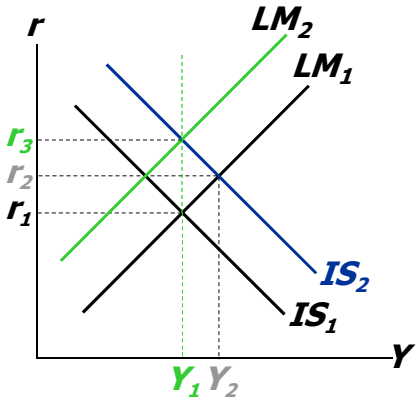
If government raises ***G***, the *IS* curve shifts right.

To keep ***Y*** constant, central bank reduces ***M*** to shift *LM* curve left.

Results:

$$\Delta Y = 0$$

$$\Delta r = r_3 - r_1$$




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# Estimates of fiscal policy multipliers

*from the DRI macroeconometric model*

| Assumption about monetary policy | Estimated value of $\Delta Y/\Delta G$ | Estimated value of $\Delta Y/\Delta T$ |
|---|---|---|
| Fed holds money supply constant | 0.60 | –0.26 |
| Fed holds nominal interest rate constant | 1.93 | –1.19 |





# Practice

Illustrate each of the following situations with a graph showing the IS – LM model, and explain what happens to the equilibrium values of the interest rate and output:

1. An increase in public spending by the government with the money supply held constant.
2. An increase in interest rates by the central bank to offset the effects of a reduction in taxation on prices.
3. An increase in the aggregate price level due to a rise in world energy prices. The central bank and the government do not react.
4. A decrease in the money supply due to inflationary pressures by the central bank in an attempt to keep the real money supply constant.
5. A decrease in public spending by the government with the interest rates held constant by the central bank.